

No Guide, No Cheat: Detecting Smart Glasses via AR Optical Signatures

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Abstract

Smart glasses are gaining increasing attention as they are expected to complement smartphones as next-generation mobile platforms, providing an Augmented Reality (AR) interface that bridges human experience and AI perception. With the goal of making smart glasses suitable for daily use, a new security risk has emerged: smart glasses can be used as cheating tools in competitions and exams when they are indistinguishable from regular glasses. This paper delves into this issue by exploring AR display mechanisms: a waveguide embedded in the lens of smart glasses transmits light from the temples of the glasses to the user's eyes. Though the waveguide looks transparent from the front view, it induces AR displayed light leakage and unique light reflection patterns in a few other viewing angles. Motivated by this observation, we propose a new smart glasses detection method that exploits violations of the brightness constancy assumption in conventional optical flow methods to identify the unusual optical phenomena caused by the waveguide structure. Our preliminary experiments with three latest released smart glasses and three commonly seen regular glasses reveal that smart glasses can be detected in a non-invasive and privacy-preserving manner. This not only helps rebuild trustworthiness for smart glasses usage but also opens up new application opportunities that leverage AR unique optical signatures.

CCS Concepts

• **Human-centered computing** → Ubiquitous and mobile computing; Mixed / augmented reality; Displays and imagers; • **Security and privacy** → Mobile platform security.

Keywords

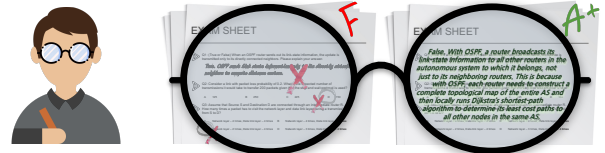
Smart Glasses, AI Glasses, AR Optics, Waveguide, Optical Flow

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1 Introduction

Smartphones have been around for decades, evolving from simple communication tools into platforms that deliver rich app experiences. However, the imminent emergence of advanced Artificial



A person wearing glasses Regular glasses view Smart glasses view

Figure 1: While smart glasses and regular glasses look identical, the wearer can clearly see AR-displayed answers to exam questions through the smart glasses.

Intelligence (AI) may open the door to a post-smartphone era. Smart glasses, built on one of the most familiar personal wearables, have the potential to complement smartphones by better understanding users' context. Not only can they deliver an Augmented Reality (AR) interface for conveying AI guidance and decision-making, but they can also perceive what users sense and interact with them anytime and anywhere. Inspired by this trend, Meta launched smart glasses such as the Meta Ray-Ban Display [10], and introduced its Orion project [9], which represents its most advanced smart glasses and is planned for future release. Google also launched Android XR to support smart glasses development in collaboration with Xreal [20]. Additionally, many other tech giants and startups, including Amazon, Snap, HTC, Magic Leap, Vuzix, INMO, Rokid, RayNeo, StarV, and Even Realities, have joined this promising market [19]. McKinsey recently estimated that the smart glasses market will exceed \$30 billion by 2030 [3].

Though exciting, smart glasses have recently attracted attention for their potential misuse with malicious intent due to their indistinguishable appearance from that of conventional eyewear. They are significantly lighter than previous AR headsets and are designed to blend in like regular glasses, making them suitable for daily wear. Therefore, many people do not notice the presence of smart glasses. Moreover, one crucial yet often overlooked aspect is: the AR content displayed on smart glasses is only *visible* to the wearer and remains *invisible* to external observers. This raises significant concerns about misuse: smart glasses are increasingly being used as cheating tools in competitions and exams. A student might preload textbook content onto the glasses, similar to the teleprompter functionality available in some smart glasses today [12], as shown in Figure 1. Recent reports show that one 17-year-old teenager customized a pair of smart glasses to display optimal moves during over-the-board games [4], while another person used them to cheat on a chemistry test [18]. Many people remain unaware that smart glasses have evolved beyond bulky goggles, making it difficult for them to distinguish whether a student is wearing regular glasses or smart glasses [13]. When the naked eye cannot tell the difference, it becomes time-consuming and impractical to ask people to remove their glasses for comprehensive inspection. In this paper, we pose



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the following research question: *Can we distinguish smart glasses from regular glasses in a non-invasive manner?*

To address this, we dig deep into the AR optical display principles of smart glasses. The light source that modulates virtual information into light is hidden in the temple of the glasses. This modulated light is then transmitted to a special optical component embedded in the lenses, *waveguide*, which confines the light along a defined optical path on the lens and delivers it to the wearer’s eyes. While the waveguide is exposed to the air, unlike the light source hidden in the temple, it is made with transparent materials. This makes external observers perceive the glasses as ordinary eyewear. Thus, the first challenge in detecting smart glasses is *how to detect the only exposed AR optical component of smart glasses when it is also transparent*.

To address this challenge, we observe that some light leaks out during its propagation along the waveguide due to its diffractive behavior. However, this light leakage is not visible from the front view, such as during eye contact in normal conversations, but instead appears only at a few specific angles when viewing the smart glasses. This phenomenon is called *eye glow*, as the leaked light overlaps with the user’s eyes from an external observer’s perspective. If the smart glasses are not activated, no light travels through the waveguide, making detection more challenging. We further observe that, because the waveguide is composed of multiple diffractive gratings, its periodic structure creates bright regions that resemble rainbows. We refer to this as the *rainbow pattern*. However, both eye glow and rainbow patterns are transient phenomena that are difficult for the human eye to detect under normal observation. One possible solution is to have external observers use off-the-shelf smartphone cameras to scan wearers from multiple angles and analyze the recorded video to detect the presence of eye glow or rainbow patterns. However, such scanning may unintentionally capture wearers’ faces, raising additional privacy concerns. This brings us to the second challenge: *how can we efficiently detect smart glasses while safeguarding the wearer’s privacy?*

To address this challenge, we propose two novel steps to detect the presence of smart glasses. In the first step, we borrow an idea from video motion estimation, which analyzes pixel differences between video frames to extract the motion trajectories of objects. A classical method in this domain is optical flow, which represents the movement direction and magnitude of pixels between frames using vector fields. The direction of pixel movement is encoded as color, and the magnitude is represented by brightness. With this distinctive transformation, original visual frames containing fine-grained facial details are converted into optical flow field representations to remove sensitive identity features.

The key to the second step lies in our observation that both eye glow and rainbow pattern induce *extra brightness* in certain video frames, compared to other frames that show no light leakage or reflection. One of the cornerstones in optical flow methods is the *brightness constancy assumption* [16], which states that the brightness of a pixel remains constant between consecutive frames as it moves. However, this foundational assumption is strongly violated when these special optical phenomena from smart glasses appear in certain frames. This results in fragmentation in the transformed optical flow: under normal conditions, the lens exhibits a uniformly colored flow (i.e., consistent direction and magnitude of movement

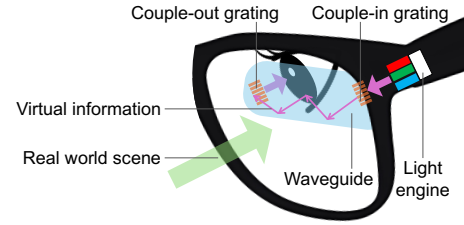


Figure 2: Illustration of the AR waveguide displays.

across the lens), but in affected frames, patches of different colors suddenly appear, indicating regions with different movement patterns than the rest of the lens. This enables us to accurately detect the presence of smart glasses across large volumes of captured video frames.

Building on this design concept, we experiment with three latest released smart glasses, Even Realities G1, StarV Air2, and Snap Spectacles, alongside three commonly used regular glasses. Preliminary findings are encouraging, demonstrating that this optical flow-based method can detect smart glasses with an average detection rate of 80%. Our exploration also highlights that existing smart glasses, such as Even Realities G1, may already possess the capability to seamlessly blend into everyday life. This opens up not only new attack vectors but also application opportunities that leverage leaked AR content from smart glasses.

2 Background

AR waveguide displays. Smart glasses employ AR waveguide technology using a set of optical combiners:

Light engine. A micro-optical projector, typically composed of a RGB MicroLED array and located within the temple of the glasses, generates input light containing virtual information, e.g., virtual objects and displayed text.

Waveguide principle. A waveguide is made of materials with different refractive indices and operates based on the principle of total internal reflection. This ensures that light remains confined within the waveguide by reflecting internally at the boundaries rather than escaping through the material, as shown in Figure 2.

Diffractive gratings. As illustrated in Figure 2, these core components are fabricated using materials whose refractive index varies periodically at the nanoscale. These dimensions are comparable to the wavelength of visible light (450–700 nm). Such periodic structures split incident light into multiple diffraction orders, each propagating in a different direction. By carefully designing parameters such as the grating’s refractive index, geometry, thickness, and duty cycle, it is possible to maximize the diffraction efficiency in a desired direction, ensuring that most of the light energy propagates along the intended path. Light enters the waveguide through a *couple-in grating*, undergoes total internal reflection during propagation, and is eventually directed toward the user’s eye through a *couple-out grating*, thereby enabling the display of AR content.

These optical combiners are typically ultra-thin (~1 mm) and lightweight (~4 g) [5]. Their high transparency to ambient light allows them to seamlessly blend virtual content with real-world scenes, providing an optical see-through AR experience.

Optical flow field. Optical flow algorithms are employed to extract motion information from consecutive video frames. Optical

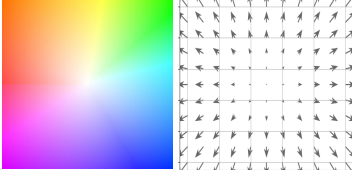


Figure 3: Illustration of the optical flow representation.

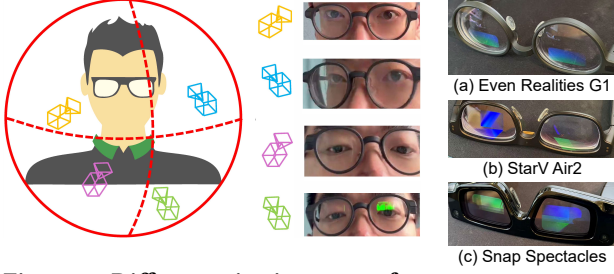


Figure 4: Different viewing poses for Figure 5: Rainbow smart glasses. Only the green view pose patterns observed successfully detects the eye glow. for 3 smart glasses.

flow represents the instantaneous motion of pixels on the image plane, resulting from the relative movement between the camera and objects in the real-world scene. By computing pixel movement between adjacent frames, one can generate an optical flow vector field, which encodes both the direction and magnitude of motion for every pixel. As shown in Figure 3, different colors and brightness levels in the optical flow representation indicate the direction and magnitude of movement, respectively.

3 Opportunities and Challenges

In this section, we describe the phenomena we observed that can be leveraged to detect smart glasses.

3.1 Eye Glow

Eye glow refers to the phenomenon where the user’s eye region appears to be surrounded by a visible glow when wearing an AR device. This is caused by light leakage from the AR-displayed content. Earlier AR headsets primarily used the birdbath display architecture, which employs a spherical reflector in conjunction with a beamsplitter to combine virtual imagery with external light. However, such systems inherently reflect a significant portion of the light emitted from the microdisplay outward, making light leakage a prominent issue in past optical see-through AR headsets. Existing studies have reported that light leakage from the HoloLens 2 can reach approximately 40% [7]. To mitigate this issue, modern smart glasses have adopted advanced waveguide-based AR optics designed to minimize AR-displayed light leakage. However, our preliminary study, conducted using a smartphone’s selfie camera to scan a user wearing smart glasses from various perspectives within the hemispherical region illustrated in Figure 4, reveals that noticeable light leakage persists, but only at a few specific viewing angles. This light leakage, which we term *eye glow on smart glasses*, arises from *originally intended to be sent to the user’s eyes, leaking*

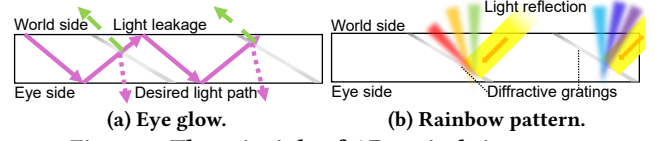


Figure 6: The principle of AR optical signatures.

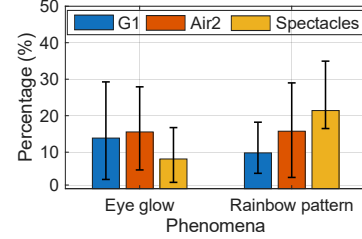


Figure 7: Percentage of eye glow and rainbow pattern phenomena observed during the entire scanning process.

partially outward toward specific angles into the environment via the diffractive gratings, as shown in Figure 6(a).

This indicates that, despite efforts to conceal forward-facing light leakage using waveguides, a non-negligible amount of light escapes from unintended directions. While this leakage remains hidden from a frontal perspective during normal social interaction, it can be observed from those rare alternative viewpoints. As shown in Figure 7, our analysis of 60 video clips from comprehensive scans of different smart glasses models reveals that the average proportion of frames exhibiting eye glow for G1, Air 2, and Spectacles is 13.87%, 15.55%, and 8.14%, respectively. Notably, in the most subtle case, a 692-frame scan of the Spectacles, eye glow was present in only 12 frames, representing a detection window of just 1.73%.

3.2 Rainbow Pattern

Detecting smart glasses solely based on active light leakage is insufficient, as smart glasses can appear nearly identical to regular glasses when a student intentionally deactivates the AR display to evade detection. To explore this further, we conduct an experiment to observe three smart glasses when they are deactivated, following the same setup shown in Figure 4. We observe a distinctive, rainbow-like streak of light appearing on the lenses, as shown in Figure 5, and we term this phenomenon the *rainbow pattern of smart glasses*. This phenomenon is caused by the *diffraction of ambient light (such as ceiling lights or natural sunlight) into multiple spectral bands*, as illustrated in Figure 6(b). Furthermore, different models of smart glasses produce varying rainbow patterns due to slight differences in their diffractive gratings, such as differences in the grating’s refractive index, geometry, thickness, and duty cycle. Nevertheless, these patterns exhibit highly similar visual characteristics across devices.

Specifically, our statistical analysis of 60 video clips reveals that the average proportion of frames containing these rainbow patterns for G1, Air 2, and Spectacles was 9.81%, 15.75%, and 21.41%, respectively. Similar to eye glow, this phenomenon is highly perspective-dependent. In the most sparse case (an Air 2 scan of 584 frames), the rainbow pattern appeared in only 18 frames, representing a detection window of just 3.08%.

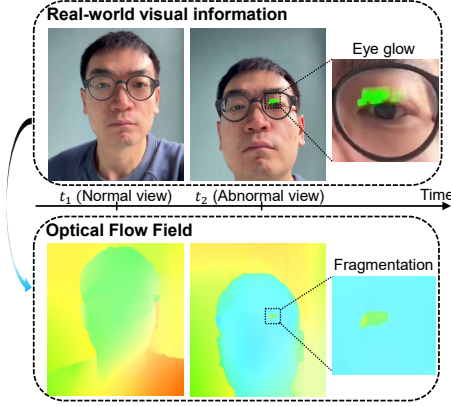


Figure 8: Illustration of the transformation to optical flow fields, as well as fragmentation in the optical flow caused by the additional brightness from eye glow.

3.3 Detection Principles

Based on the aforementioned observations, we establish the following design principles for a smart glasses detection system:

- (1) The detection system should observe the glasses from a wide range of perspectives, similar to a Face ID [1] enrollment process;
- (2) A distinguishing feature of smart glasses is that their lens regions may exhibit increased brightness caused by *eye glow* or *rainbow patterns* during scanning.

For (1), a straightforward method is to use a smartphone’s front-facing camera to scan a user wearing glasses, ensuring a pair of glasses is observed from a comprehensive range of angles. However, Face ID builds a 3D facial model using structured light without capturing any actual visual facial data. In contrast, scanning with a regular front camera captures the wearer’s key private attributes, such as visual facial information. This motivates the need for a privacy-preserving approach to smart glasses detection. For (2), the brightness changes in the lens regions are often transient, appearing in only a few frames out of hundreds, and are nearly imperceptible to the naked eye. Yet, detecting these subtle temporal changes is critical to confirming the presence of smart glasses. Consequently, we propose a novel detection pipeline designed specifically for this purpose. Next, we present the details.

4 Smart Glasses Detection Pipeline

4.1 Optical Flow Transformation

Since optical flow uses pixel motion differentiation to eliminate the fine-grained facial textures, it serves as an effective mechanism for converting raw visual frames into a two-dimensional optical flow vector field with different colors and brightness levels, as shown in Figure 8. However, processing the entire selfie image would impose a significant computational burden on the detector. Furthermore, only the glasses region constitutes our Region of Interest (RoI). To address this, we isolate the glasses within the optical flow field using the Segment Anything Model (SAM) [11]. As illustrated in Figure 9, glasses masks are generated for frames captured from various perspectives. Both the optical flow transformation and SAM

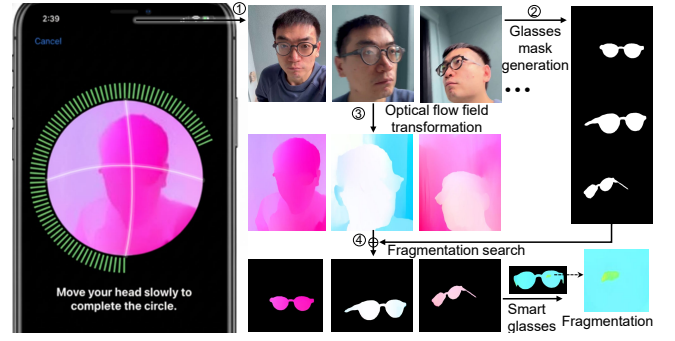


Figure 9: Illustration of smart glasses detection pipeline.

mask generation are executed within the smartphone’s protected execution environment, and the raw visual frames are purged immediately after processing. Consequently, only the anonymized optical flow fields and glasses masks, both of which exclude sensitive biometric attributes, are transmitted to the edge server for further analysis.

4.2 Brightness Constancy Violation

A fundamental assumption in the optical flow algorithms is *brightness constancy*: the brightness of a specific pixel in a scene remains constant across consecutive frames, despite its motion. This assumption leads to the basic formulation of optical flow. Specifically, consider a pixel at position (x, y) in frame t that moves to a new position $(x + \Delta x, y + \Delta y)$ over a temporal interval Δt , the brightness constancy assumption states that:

$$\underbrace{I(x, y, t)}_{\text{Brightness at } t} = \underbrace{I(x + \Delta x, y + \Delta y, t + \Delta t)}_{\text{Brightness at } t + \Delta t} \quad (1)$$

By applying a first-order Taylor expansion, the optical flow vector $[u, v]^T = [\Delta x / \Delta t, \Delta y / \Delta t]^T$ can be derived as:

$$\begin{bmatrix} I'_x & I'_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = -I'_t \Delta t, \quad (2)$$

where $I'_x = \partial I / \partial x$, $I'_y = \partial I / \partial y$, and $I'_t = \partial I / \partial t$ represent the spatial (x, y) and temporal (t) derivatives of the brightness of the image. These partial derivatives can be estimated from the input visual image sequence to solve the optical flow vector $[u, v]^T$.

However, if transient brightness fluctuations, caused by the emergence or disappearance of eye glow or rainbow patterns, occur within a few frames, the brightness constancy assumption is violated. In such cases, the standard constraint equations no longer hold and must be replaced by expressions that account for brightness residuals:

$$\underbrace{\begin{bmatrix} I'_x & I'_y \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}}_{\text{Brightness}\uparrow} = \underbrace{-\delta(x, y) - I'_t \Delta t}_{\text{Brightness}\downarrow}, \quad (3)$$

where $\delta(x, y)$ represents the local brightness variation at position (x, y) . These brightness changes manifest as fragmentation within the optical flow field (See Figure 8), where patches of different colors suddenly appear within what would otherwise be a uniformly colored optical flow. It is important to note that our use of the optical flow method differs from conventional motion estimation. While typical scenarios aim to extract object trajectories, our approach

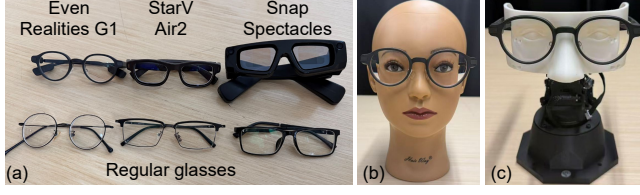


Figure 10: Setup: (a) 3 smart and 3 regular glasses (Numbered 1-3 from left to right); (b) Mannequin head wearing G1; (c) 3D-printed face mask wearing G1 held by a robot arm.

assumes the motion of the glasses is already known. Specifically, the glasses’ trajectory and masks are pre-obtained via SAM. We focus instead on analyzing these fragmentation patterns within the isolated glasses region to detect the presence of smart glasses. To achieve this, we employ RAFT [17], a neural network-based optical flow estimation model, to capture these subtle flow fragmentations. Lastly, we apply K-means clustering based on the silhouette score. If two or more distinct clusters appear within the optical flow region of the glasses, it indicates that fragmentation has occurred, thereby signaling the presence of smart glasses.

5 Performance Evaluation

5.1 Evaluation Setup

We employ three off-the-shelf smart glasses as target devices, *Even Realities G1*, *StarV Air2*, and *Snap Spectacles*, as shown in Figure 10(a). Each of these devices incorporates a transparent waveguide to overlay AR content onto the real-world scene. For the reference group, we randomly select three pairs of regular prescription glasses (regular-1 at -8.50 D, and regular-2 and regular-3 at -4.50 D). An iPhone 17 Pro serves as the detection device. To ensure consistency, we mount all glasses on two types of facial models: a *mannequin head* and a *custom 3D-printed face mask*. Following the procedure outlined in Figure 9, we scan each model for approximately 20 seconds. We capture 30 pairs of 720p video clips per model, totaling 60 video clips per pair of glasses. The captured data is transmitted to a PC equipped with an NVIDIA RTX 4090 GPU and 64 GB of RAM. The proposed detection pipeline is implemented in Python using PyTorch. We use SAM 2.1 [11] with the small model weights configuration. The RAFT model [17], used for optical flow field transformation, is trained on the standard FlyingChairs dataset [6]. We evaluate the system using two primary metrics: (1) *Detection rate*: A successful detection of smart glasses occurs when the system correctly identifies the target glasses as smart glasses. The detection rate should ideally be 100%; (2) *False positive rate*: A false positive occurs when the system incorrectly identifies regular glasses as smart glasses. The ideal target for the false positive rate is 0%.

5.2 Preliminary Results

We evaluate the detection pipeline across both facial models and all target glasses, with the results summarized in Figure 11. The detection rate exceeds 80% for the Even Realities G1 (83.3% on the 3D-printed face mask and 96.7% on the mannequin head model). However, performance is lower for the other two smart glasses, with Snap Spectacles achieving a detection rate of approximately 60%. This disparity is attributed to the fact that the Spectacles exhibit fewer angles of observable light leakage compared to the other

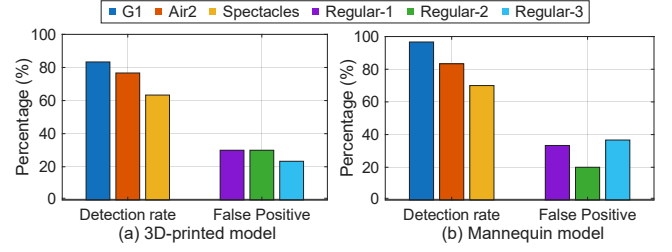


Figure 11: Performance across different glasses.

two devices, as shown in Figure 7. Overall, detection accuracy is consistently higher on the mannequin head than on the 3D-printed mask. This is because SAM’s glasses mask generation is more robust on the mannequin head, whereas the 3D-printed face is not a complete head model, which occasionally leads to inaccurate SAM-generated masks, resulting in incorrect selection of the RoI regions. These findings suggest that future evaluations should utilize facial models that more closely approximate realistic human anatomy. Furthermore, leveraging more advanced semantic segmentation, such as SAM 3 [2], could yield more precise masks and improve the reliability of the results. On the other hand, the detection rate for regular glasses was approximately 25% on both facial models.

6 Discussion and Future Directions

Analysis of fragmentation and brightness patterns. The current detection pipeline still suffers from a relatively high false positive rate, primarily due to the glossy nature of glasses, which can reflect ambient light sources such as ceiling lights. These reflections from regular glasses also violate the brightness constancy assumption. One possible solution is to build a spectral feature database that captures the eye glow and rainbow patterns of a wide range of smart glasses. However, the rapid evolution of smart glasses models, combined with user-customized designs, poses substantial challenges in constructing such a comprehensive feature database. A more promising direction may involve leveraging the powerful reasoning capabilities of Vision-Language Models (VLMs) to analyze the fragmentation patterns and corresponding brightness features in the detected region, determining whether they result from eye glow, rainbow patterns, or simple specular reflections from the environment. In the case of eye glow, VLMs may even be capable of reconstructing the original AR content from blurred or leaked displays, raising potential concerns about AR content privacy and trust, which we will discuss later. Compared to directly uploading entire videos to VLMs, which incurs significant time and token overhead, it would be more efficient and practical to upload only filtered keyframes for VLM-assisted decision-making.

Human-centered detector design. Currently, the scanning process is performed manually, which introduces variability in the scanning trajectories and may lead to inconsistent detection performance in real-world deployments. To address this, we mount the 3D-printed face mask on a robotic arm (the PincherX 100 from Trossen Robotics), and we plan to use ROS programming to mimic various user scanning behaviors and study their impact on detection accuracy in future evaluations. To improve the end-user experience, we aim to develop an indicator UI, similar to the current Face ID enrollment process (see Figure 9). This UI will guide users during scanning, ensuring that all viewing angles of the smart glasses are

covered when using the detector app. In parallel, we plan to conduct a user study to investigate whether users can identify the presence of smart glasses in a non-invasive manner (i.e., using only their eyes and without physically touching the glasses) and to evaluate the overall utility of our smart glasses detector.

Multi-modal detection. As a core aspect of the smart glasses misuse problem in cheating scenarios, this work focuses solely on detecting waveguide-based displays. Another key output component, the speaker, is not considered, as traditional acoustic signals are easily noticeable by external observers, making them less suitable for covert use. However, the emergence of new bone-conduction speaker technologies may allow output that is inaudible to others. Detecting such signals poses an interesting new challenge. In addition, once activated, smart glasses may emit side-channel signals that can potentially be detected, such as RF signals [8, 15] or thermal emissions [22]. These signals often have long propagation ranges, which could inspire the development of long-distance smart glasses detection techniques. Beyond output components, input components, such as cameras and microphones, could also be misused or compromised, especially if miniature cameras (e.g., ~1 mm) are eventually integrated into smart glasses and deployed in privacy-sensitive environments such as bathrooms. Given the uniquely sensitive nature of wearable devices, whether existing camera detection methods [14, 21] can be adapted to detect cameras embedded in smart glasses remains an open question.

Trustworthiness of AR content on smart glasses. AR users inherently trust that AR content is visible only to themselves, and thus such content is assumed to be private (e.g., a concealed teleprompter displayed during public speaking). AR overlays on smart glasses can provide vastly different experiences of the same real-world scene to different users. When this private content is exposed to others through unintended viewing angles, rebuilding public trust in AR content displayed on smart glasses becomes a critical challenge. A fundamental question is whether it is even possible to design a perfect AR waveguide system: one that allows light to travel from the glasses' temple arms, through the waveguide to reach the user's eyes with zero loss. Snap Spectacles exhibit a lower light leakage rate (i.e., fewer instances of eye glow), but they introduce more frequent rainbow patterns caused by environmental light reflections. Therefore, the design and selection of waveguide technology will play a key role in reestablishing user trust in the AR content displayed on smart glasses. More broadly, it is also worth exploring whether this inside-out visibility (i.e., projecting content outward toward the public) could enable new application opportunities, beyond the traditional mode where virtual content is designed only for AR users.

7 Conclusion

While smart glasses open new opportunities for interactive, embodied, and context-aware AI systems, there is also growing concern about their potential misuse for malicious purposes, such as cheating on exams. To mitigate the risks associated with smart glasses, we examined AR waveguide display mechanisms and identified two unique AR optical signatures: eye glow and rainbow patterns. We further observed that these AR signatures can violate the brightness constancy assumption in optical flow fields. Building upon this, we developed a non-invasive smart glasses detection pipeline,

evaluated on three types of smart glasses and three pairs of regular glasses, achieving a detection rate of over 80%. Beyond auditing the use of smart glasses in sensitive settings, our findings illuminate a path to leverage AR's unique optical signatures to inform new smart glasses designs and enable applications beyond today's wearer-only display paradigm.

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